

Information and Estimation Theory
Lecture 08: Rate-distortion theory

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Rate and Distortion

- A real number requires an infinite number of bits, so a finite representation of a continuous RV can never be perfect
- How well can we do?
  - To answer the question appropriately, it is necessary to define the "goodness" of a representation of a source
- Define a distortion measure, which is a measure of distance between the RV and its representation
- The basic problem in **rate-distortion (RD)** theory can then be stated as follows
  - Given a source distribution and a distortion measure, what is the minimum expected distortion achievable at a particular rate?
  - (Equivalently) What is the minimum rate description required to achieve a particular distortion?

Rate and Distortion

- One of the most intriguing aspects of this theory is that joint descriptions are more efficient than individual descriptions
  - It is simpler to describe an elephant () and a chicken () with one description than to describe each alone
- It is simpler to describe 2 independent RVs X_1 and X_2 together (at a given distortion) than to describe each by itself
- Why don't independent problems have independent solutions?
  - The answer is Geometry: rectangular grid points (arising from independent descriptions) do not fill up the space efficiently
- RD theory can be applied to both discrete and continuous RVs

- We begin by considering the simple problem of representing a single continuous RV by a finite number of bits

Quantization

- A continuous random source requires ∞ precision to represent, we cannot reproduce it exactly using a finite-rate code
- Find the best possible representation for any given data rate

- Consider representing a single sample from the source, X, a RV, with its representation $\hat{X}(X)$
- With R bits to represent X, the function $\hat{X}$ can take on 2^R values
- Find the optimum set of values for $\hat{X}$ (called the **reproduction points** or **code points**) and the **representation regions** that are associated with each value $\hat{X}$

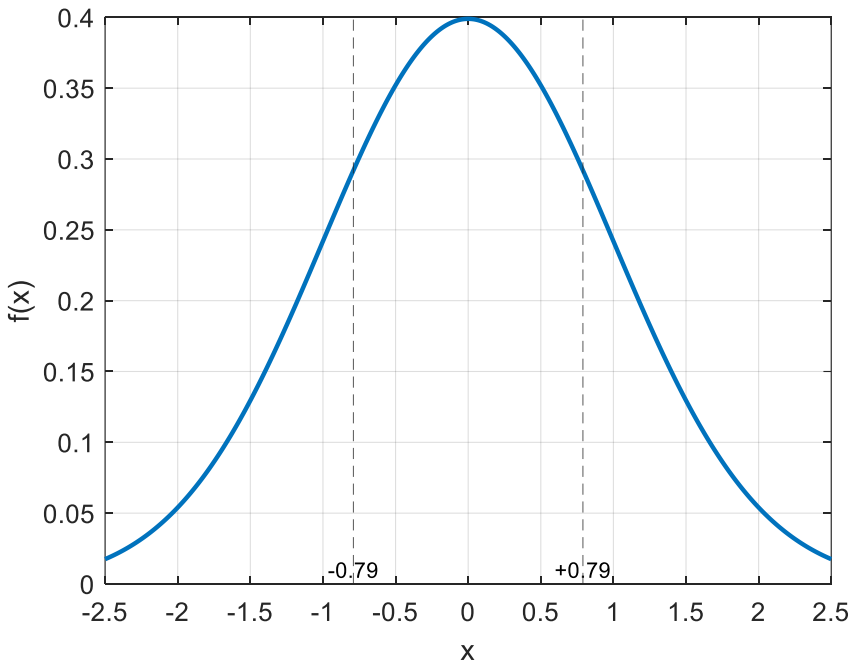
Gaussian with 1 Bit

- E.g., let $X \sim \mathcal{N}(0,\sigma^2)$, and a squared-error distortion measure
- We wish to find the function $\hat{X}(X)$ such that $\hat{X}$ takes on at most 2^R values and minimizes the **squared error** $E\{(X-\hat{X}(X))^2\}$
- With 1 bit to represent X, the bit should distinguish whether $X>0$
- To minimize the squared error, each **reconstruction point** should be the **conditional mean** of its **representation region**

E{x|x≥0}

$$\hat{X}(x) = \begin{cases} \sqrt{\frac{2}{\pi}}\sigma, & x \geq 0 \\ -\sqrt{\frac{2}{\pi}}\sigma, & x < 0 \end{cases}$$

E{x|x<0}



Gaussian with 1 Bit

- *Proof*: Find $\min_{T,x_0,x_1} \mathrm{E}\left\{\left(X-\hat{X}\right)^2\right\}$
- Let $\hat{X} \in \{x_0, x_1\}$ and T be the boundary btw. the two representation regions

d/dy $\int_{-\infty}^{h(y)} g(x)dx = g(h(y))h'(y)$

$$\frac{\partial}{\partial T}\mathrm{E}\left\{\left(X-\hat{X}\right)^2\right\} = \frac{\partial}{\partial T}\left(\int_{-\infty}^{T}(x-x_0)^2 f(x)\,dx + \int_{T}^{+\infty}(x-x_1)^2 f(x)\,dx\right)$$

$$= (T-x_0)^2 f(T) - (T-x_1)^2 f(T) = -2Tx_0 + 2Tx_1 + x_0^2 - x_1^2 = 0 \quad f(T)>0$$

$$-2T(x_0-x_1) + (x_0+x_1)(x_0-x_1) = 0 \quad x_0<x_1 \qquad T = \frac{x_0+x_1}{2}$$

$$\frac{\partial}{\partial x_0}\mathrm{E}\left\{\left(X-\hat{X}\right)^2\right\} = \frac{\partial}{\partial x_0}\left(\int_{-\infty}^{\frac{x_0+x_1}{2}}(x-x_0)^2 f(x)\,dx + \int_{\frac{x_0+x_1}{2}}^{+\infty}(x-x_1)^2 f(x)\,dx\right)$$

Regola della composizione

$$= (T-x_0)^2 f(T)\frac{1}{2} + \int_{-\infty}^{\frac{x_0+x_1}{2}} \frac{\partial}{\partial x_0}(x-x_0)^2 f(x)dx - (T-x_1)^2 f(T)\frac{1}{2} = 0$$

$$\int_{-\infty}^{T} -2(x-x_0)f(x)dx = 0. \qquad \int_{-\infty}^{T} xf(x)dx - x_0\int_{-\infty}^{T} f(x)dx = 0$$

$x_0 = \frac{\int_{-\infty}^{T} xf(x)dx}{\int_{-\infty}^{T} f(x)dx}$ and similarly $x_1 = \frac{\int_{T}^{\infty} xf(x)dx}{\int_{T}^{\infty} f(x)dx}$.

Gaussian with 1 Bit

- Due to the even symmetry of $f(x)$, $x_1 = -x_0$
- So, T=0 and the centroids are

$$x_0 = \frac{\int_{-\infty}^{0} x f(x)dx}{\int_{-\infty}^{0} f(x)dx} = \mathrm{E}\{X|X<0\} = \frac{\frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{0} x e^{-\frac{x^2}{2\sigma^2}}dx}{\frac{1}{2}} = \frac{2}{\sqrt{2\pi}\sigma}\left[-\sigma^2 e^{-\frac{x^2}{2\sigma^2}}\right]_{-\infty}^{0}$$

$$= \frac{2\sigma}{\sqrt{2\pi}}(-1) = -\sqrt{\frac{2}{\pi}}\sigma \qquad x_1 = \sqrt{\frac{2}{\pi}}\sigma$$

- The distortion is

$$\mathrm{E}\left\{\left(X-\hat{X}\right)^2\right\} = \mathrm{E}\{X^2\} - 2\mathrm{E}\{X\hat{X}\} + \mathrm{E}\{\hat{X}^2\}$$

$$= \sigma^2 - 2\frac{1}{2}\mathrm{E}\{X\hat{X}|X<0\} - 2\frac{1}{2}\mathrm{E}\{X\hat{X}|X\geq 0\} + \frac{1}{2}x_0^2 + \frac{1}{2}x_1^2$$

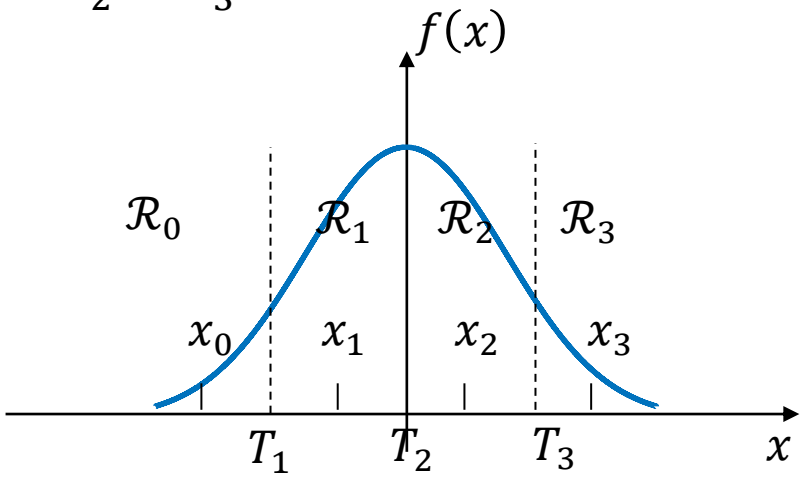
$$= \sigma^2 - x_0\mathrm{E}\{X|X<0\} - x_1\mathrm{E}\{X|X\geq 0\} + x_0^2$$

$$= \sigma^2 - x_0^2 - x_1^2 + x_0^2 = \sigma^2 - x_1^2 = \sigma^2 - \frac{2}{\pi}\sigma^2 = \frac{\pi-2}{\pi}\sigma^2$$

$$= 0.363\sigma^2$$

Gaussian with 2 Bits

- With 2 bits per sample, the situation is not as simple
- Clearly, we want to divide the real line into four representation regions
- Each region is associated to a reconstruction point that represents the original RV sample
- For a Gaussian RV and MSE distortion
  - Choose 3 region boundaries $T_1 < T_2 < T_3$
    - Four regions $\mathcal{R}_0 = (-\infty, T_1],\ \mathcal{R}_1 = (T_1, T_2], \mathcal{R}_2 = (T_2, T_3], \mathcal{R}_3 = (T_3, +\infty]$
  - Choose 4 reconstruction points $x_0 < x_1 < x_2 < x_3$
- For a zero-mean Gaussian RV, even symmetry results in a symmetric optimal quantizer
  - $T_1 = -T_3, T_2 = 0$
  - $x_0 = -x_3, x_1 = -x_2$



Nearest-Neighbor/Centroid Conditions

- Two simple properties of optimal regions and reconstruction points characterize the optimal quantization of a single RV (see 1-bit Gaussian) under MSE distortion

  - Given a set of reconstruction points $\{\hat{X}\}$, the MSE distortion is minimized by mapping a source RV X to its closest representation $\hat{X}$, i.e., its **nearest neighbor**
    - The set of regions of X is called a **Voronoi** or **Dirichlet partition**

$$T_i = \frac{x_{i-1} + x_i}{2}$$

  - Given the representation regions, the reconstruction points $\{\hat{X}\}$ should minimize the conditional MSE distortion over their respective assignment regions $\mathcal{R}_i$, i.e., it is the **conditional mean** or **centroid**

$$x_i = \mathrm{E}\{X | T_{i-1} \leq X < T_i\} = \mathrm{E}\{X | X \in \mathcal{R}_i\}$$

Lloyd-Max Algorithm

- The two properties enable us to construct a simple algorithm to find a "good" quantizer
  - We start with a set of reconstruction points $\{\hat{X}\}$
  - We find the optimal set of reconstruction regions (which are the nearest-neighbor regions with respect to the distortion measure)
  - We find the optimal reconstruction points for these regions (the centroids of these regions if the distortion is squared error)
  - Repeat the iteration for this new set of reconstruction points

- The expected distortion is decreased at each stage in the algorithm, so it will converge to a local minimum of the distortion
- This algorithm is called the **Lloyd-Max** (single RV) or **generalized Lloyd algorithm** (vector RV) and is frequently used to design quantization systems

Example: Lloyd-Max on 2-Bit Gaussian

- Zero-mean Gaussian with unit variance
  - $-T_1 = T_3 = T, T_2 = 0$
  - $-x_0 = x_3 = \alpha, -x_1 = x_2 = \beta, 0 < \beta < \alpha$
- Lloyd-Max properties are <u>numerically</u> and <u>iteratively</u> evaluated
  - $\alpha^{(k)} = \mathrm{E}\{X | X > T^{(k-1)}\} = \frac{\phi(T^{(k-1)})}{Q(T^{(k-1)})},$

    $\int x\phi(x)dx = -\phi(x) + C$
    Q(x) is the Gaussian tail function
  - $\beta^{(k)} = \mathrm{E}\{X | 0 < X \leq T^{(k-1)}\} = \frac{\phi(0) - \phi(T^{(k-1)})}{Q(0) - Q(T^{(k-1)})} = \frac{(2\pi)^{-\frac{1}{2}} - \phi(T^{(k-1)})}{\frac{1}{2} - Q(T^{(k-1)})}$
  - $T^{(k)} = \frac{\alpha^{(k)} + \beta^{(k)}}{2}$
- With $T^{(0)} = 1$, we obtain the sequence $(k, \alpha^{(k)}, \beta^{(k)}, T^{(k)})$
  - 1 1.5251 0.4599 0.9925
  - 2 1.5191 0.4570 0.9881
  - 3 1.5156 0.4553 0.9854
  - …
  - 100 1.5104 0.4528 0.9816

$$x_0 = -1.5104$$
$$x_1 = -0.4528$$
$$x_2 = +0.4528$$
$$x_3 = +1.5104$$

$$T_1 = -0.9816$$
$$T_2 = 0$$
$$T_3 = +0.9816$$

$$\mathrm{E}\left\{\left(X - \hat{X}\right)^2\right\} = 0.1175$$

Improving the Idea

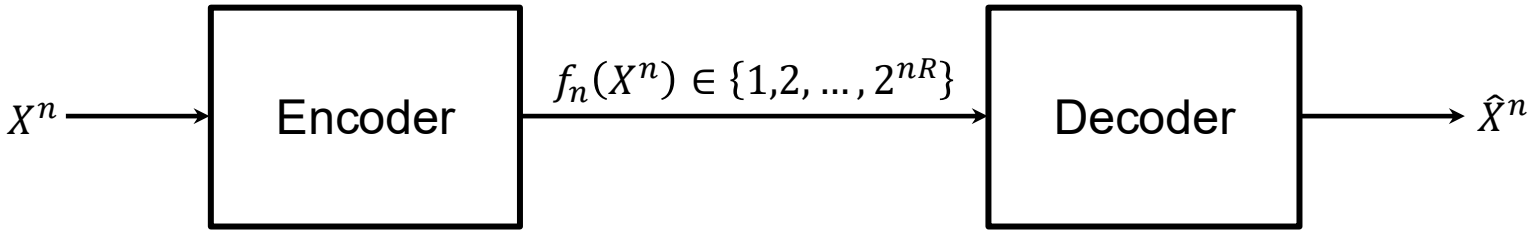
- Instead of quantizing a single RV, use a set of *n* RVs, e.g., each RV i.i.d.$\sim\mathcal{N}(0,\sigma^2)$
  - These RVs are represented by *nR* bits

- Since the source is i.i.d., the symbols are independent, and it may appear that the representation of each element is an independent problem to be treated separately
  - **This is generally not true**, as the later results on RD theory will show

- We represent the whole sequence by a single index taking 2^{nR} values
- This treatment of entire sequences at once achieves a lower distortion for the same rate than independent quantization of the individual samples

Definitions

- Let us formalize and generalize the previous findings
- Assume that we have a source that produces a sequence $X_1, X_2, \ldots, X_n$ i.i.d. $\sim p(x)$, $x \in \mathcal{X}$
  - We use a finite alphabet, but this can be extended to continuous RV
- The encoder describes the source sequence X^n by an index $f_n(X^n) \in \{1,2,\ldots,2^{nR}\}$
- The decoder represents X^n by an estimate $\hat{X}^n \in \hat{\mathcal{X}}^n$



- At rate R, a block of length n is represented by one of 2^{nR} indices

Distortion

- A **distortion function** or **distortion measure** is a mapping

$$d: \mathcal{X} \times \hat{\mathcal{X}} \rightarrow \mathbb{R}^+$$

Takes as inputs the original and the decoded sequences and produces a real nonnegative number

from the set of source alphabet-reproduction alphabet pairs into the set of nonnegative real numbers

- The distortion ***d*(*x*,*x̂*)** is a measure of the cost of representing the symbol *x* by the symbol $\hat{x}$

- A distortion measure is said to be **bounded** if the maximum value of the distortion is finite

$$d_{\max} \triangleq \max_{x \in \mathcal{X}, \hat{x} \in \hat{\mathcal{X}}} d(x, \hat{x}) < \infty$$

- In most cases, the reproduction alphabet $\hat{\mathcal{X}}$ is the same as the source alphabet $\mathcal{X}$

Hamming Distortion (Probability of Error)

- The **Hamming distortion** (bounded) is given by

$$d(x,\hat{x}) = \begin{cases} 0, & x = \hat{x} \\ 1, & x \neq \hat{x} \end{cases}$$

  which results in a **probability of error distortion**

- The average Hamming distortion results in the probability of error

$$E\{d(X,\hat{X})\} = 0 \cdot \Pr\{X = \hat{X}\} + 1 \cdot \Pr\{X \neq \hat{X}\} = \Pr\{X \neq \hat{X}\}$$

- Example
  - 8% symbol errors ⇒ distortion = 0.08

Squared-Error Distortion

- **Mean-squared error (MSE)** for continuous alphabets (also unbounded)

$$d(x, \hat{x}) = (x - \hat{x})^2$$

Also check SNR & PSNR

- Popular, simple, and related to least-squares prediction
- Sometimes MSE is not appropriate for human "consumers"
  - E.g., large MSE between a speech and its slightly time-shifted copy, but they sound the same to a human listener
  - Wang & Bovik (2004) conceived the structural similarity index (SSIM): all boat images below have MSE=210, but SSIM increases from left to right



Distortion of Sequences

- The distortion between sequences x^n and $\hat{x}^n$ is defined by

$$d(x^n, \hat{x}^n) = \frac{1}{n}\sum_{i=1}^{n} d(x_i, \hat{x}_i)$$

- So, the distortion between two sequences is the average of the per-symbol distortion of their elements
- This is not the only reasonable definition
  - E.g., measure the distortion between two sequences by the maximum or the sum of the per-symbol distortions

$$d_{\mathrm{MAX}}(x^n, \hat{x}^n) = \max_i d(x_i, \hat{x}_i) \qquad d_{\mathrm{SUM}}(x^n, \hat{x}^n) = \sum_{i=1}^{n} d(x_i, \hat{x}_i)$$

  - The theory we are going to develop does not apply directly to these more general distortion measures

RD Code

- A **$(2^{nR},n)$-rate distortion code** consists of an encoding function

$$f_n: \mathcal{X}^n \rightarrow \{1,2,\ldots,2^{nR}\}$$

and a decoding (reproduction) function

$$g_n: \{1,2,\ldots,2^{nR}\} \rightarrow \hat{\mathcal{X}}^n$$

- The distortion associated with the $(2^{nR},n)$ RD code is defined as

$$D = \mathrm{E}\left\{d\left(X^n, g_n\left(f_n(X^n)\right)\right)\right\} = \sum_{x^n} p(x^n) d\left(x^n, g_n\left(f_n(x^n)\right)\right)$$

where the expectation is taken with respect to the probability distribution on X^n

RD Code

- The set of n-tuples $g_n(1), g_n(2), \ldots, g_n(2^{nR})$ denoted by $\hat{X}^n(1), \hat{X}^n(2), \ldots, \hat{X}^n(2^{nR})$ constitutes the **codebook**

- $f_n^{-1}(1), \ldots, f_n^{-1}(2^{nR})$ are the associated **assignment regions**

Partition

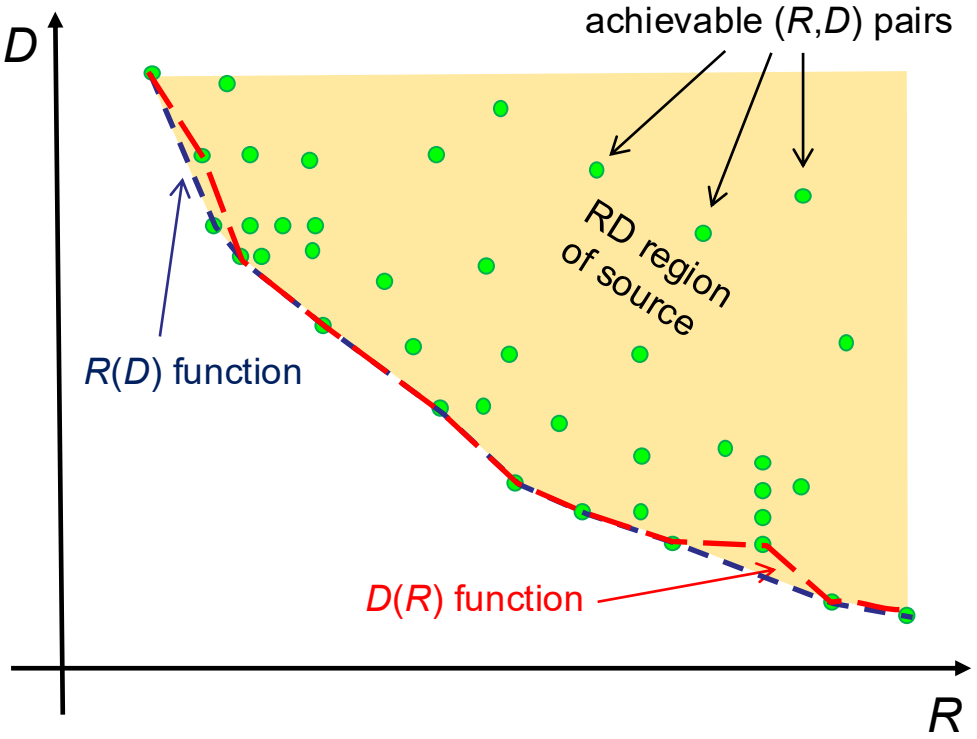
- Many terms are used to describe the replacement of X^n by its quantized version $\hat{X}^n$
- It is common to refer to $\hat{X}^n$ as the **vector quantization**, reproduction, reconstruction, representation, **source code**, or estimate of X^n

RD Regions and Functions

- A RD pair (R,D) is said to be **achievable** if there exists a sequence of $(2^{nR},n)$ RD codes (f_n,g_n) with

$$\lim_{n\to\infty} \mathrm{E}\left\{d\left(X^n, g_n\left(f_n(X^n)\right)\right)\right\} \le D$$

- The **RD region** for a source is the topological *closure* "chiusura" of the set of achievable RD pairs (R,D)
- The **RD function $R(D)$** is the *infimum* of rates R such that (R,D) is in the RD region of the source for a given D



- The **distortion rate (DR) function $D(R)$** is the *infimum* of all distortions D such that (R,D) is in the RD region for a given R

Shannon Information RD Function

- The distortion rate function defines another way of looking at the boundary of the RD region
- We will in general use the $R(D)$ function rather than the $D(R)$ function for this boundary: the two approaches are equivalent
- The previous $R(D)$ and $D(R)$ curves are **operational**, since they are defined by actual source codes

- The **information rate distortion function** $R^{(I)}(D)$ for a source X with distortion measure $d(x,\hat{x})$ is defined as

$$R^{(I)}(D) = \min_{p(\hat{x}|x):\sum_{(x,\hat{x})} p(x)p(\hat{x}|x)d(x,\hat{x}) \le D} I\left(X;\hat{X}\right)$$

For all quantization methods with bounded distortion

where the minimization is over all conditional PMFs $p(\hat{x}|x)$ for which the joint PMF $p(x,\hat{x}) = p(x)p(\hat{x}|x)$ satisfies the expected distortion constraint

RD Theorem

- **Theorem:** The RD function for an i.i.d. source $X \sim p(x)$ with bounded distortion function $d(x,\hat{x})$ is equal to the associated information RD function. The **minimum achievable rate** for distortion at most D is

Information RD

$$R(D) = R^{(I)}(D) = \min_{p(\hat{x}|x):\sum_{(x,\hat{x})} p(x)p(\hat{x}|x)d(x,\hat{x}) \le D} I(X;\hat{X})$$

Operational RD

- This theorem guarantees that the operational definition of the RD function is equal to the information definition
  - We will use $R(D)$ to denote both definitions of the RD function
- In simple words: *we cannot achieve a distortion lower than D if we describe X at a rate lower than R(D)*
  - The RD function $R(D)$ is a nonincreasing convex function of D

RD of Binary Source

- **Theorem:** The RD function for a Bern(p) source with Hamming distortion is given by

$$R(D) = \begin{cases} H_{\mathrm{B}}(p) - H_{\mathrm{B}}(D), & 0 \le D \le \min\{p, 1-p\} \\ 0, & D > \min\{p, 1-p\} \end{cases}$$

p=Pr{X=1}

  - *Proof:* Consider a binary source $X \sim$ Bern(p) with a Hamming distortion measure. Without loss of generality, we may assume that $p \le 1/2$. We wish to calculate the rate distortion function,

$$R(D) = \min_{p(\hat{x}|x):\sum_{(x,\hat{x})} p(x)p(\hat{x}|x)d(x,\hat{x}) \le D} I(X;\hat{X})$$

  - Rather than minimizing $I(X;\hat{X})$ directly, we first derive a lower bound and then show that it is achievable

RD of Binary Source (proof cont'd)

- For any joint pdf satisfying the distortion constraint, we have

$$I(X;\hat{X}) = H(X) - H(X|\hat{X}) = H_B(p) - H(X \oplus \hat{X}|\hat{X}) \geq H_B(p) - H(X \oplus \hat{X})$$

$H()$ of binary

Conditioning reduces entropy (minus=increases)

$$\geq H_B(p) - H_B(D)$$

For $D<½$, H_B increases and $\Pr\{X \neq \hat{X}\} \leq D$

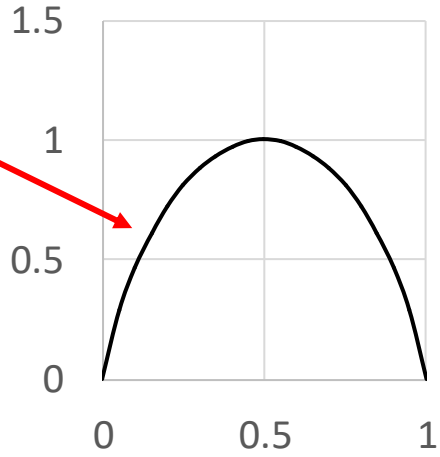
Adding a known value does not change uncertainty
Modulo-2 $X = X \oplus \hat{X} \oplus \hat{X}$

$H(X \oplus \hat{X}) = H(\Pr\{X \oplus \hat{X} = 1\}) = H(\Pr\{X \neq \hat{X}\})$

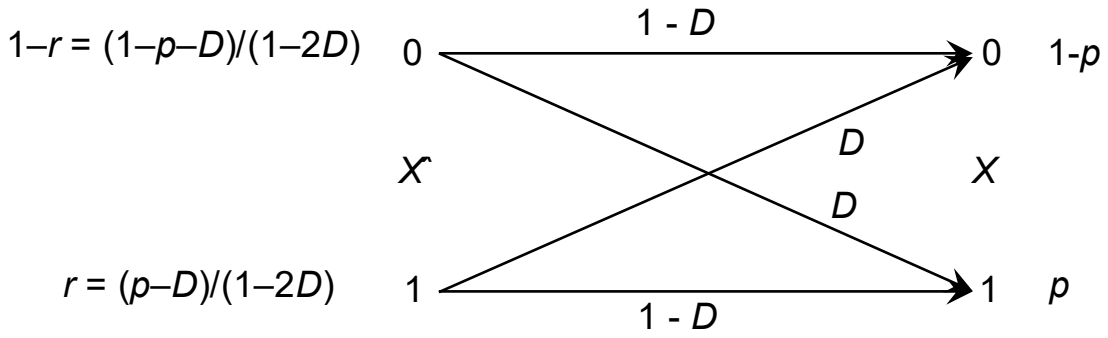
$$R(D) \geq H_B(p) - H_B(D)$$

$R(D)$ is > than its min

since $\Pr\{X \neq \hat{X}\} \leq D$ and $H(D)$ increases with D for $D \leq 1/2$



- We show that the lower bound is the RD function by finding a joint PMF that meets the distortion constraint and has $I(X;\hat{X}) = R(D)$
- For $0 \leq D \leq p$, we can achieve the value of the rate distortion function by choosing $(X,\hat{X})$ to have the joint PMF of this particular BSC



RD of Binary Source (proof cont'd)

- We choose the distribution of $\hat{X}$ at the input of the channel so that the output distribution of X is the specified distribution
- Let $r = \Pr\{\hat{X} = 1\}$ so that $r(1 - D) + (1 - r)D = p$ $\longrightarrow$ $r = \dfrac{p - D}{1 - 2D}$
- If $D \le p \le 1/2$, then $r = \Pr\{\hat{X} = 1\} \ge 0$ and $1\text{-}r = \Pr\{\hat{X} = 0\} \ge 0$

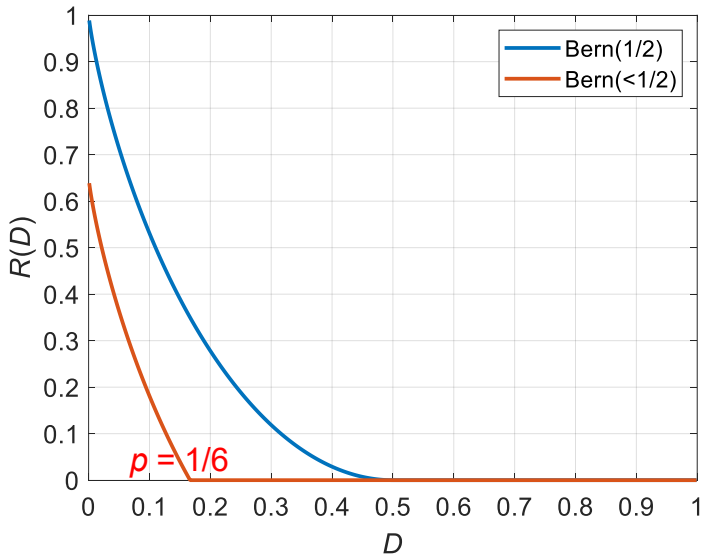
$$I(X;\hat{X}) = H(X) - H(X|\hat{X}) = H_{\mathrm{B}}(p) - H_{\mathrm{B}}(D) \longrightarrow \Pr\{X \neq \hat{X}\} = D$$

Uncertainty that there is an error

Due to high distortion, nearly all 1's are changed to 0's

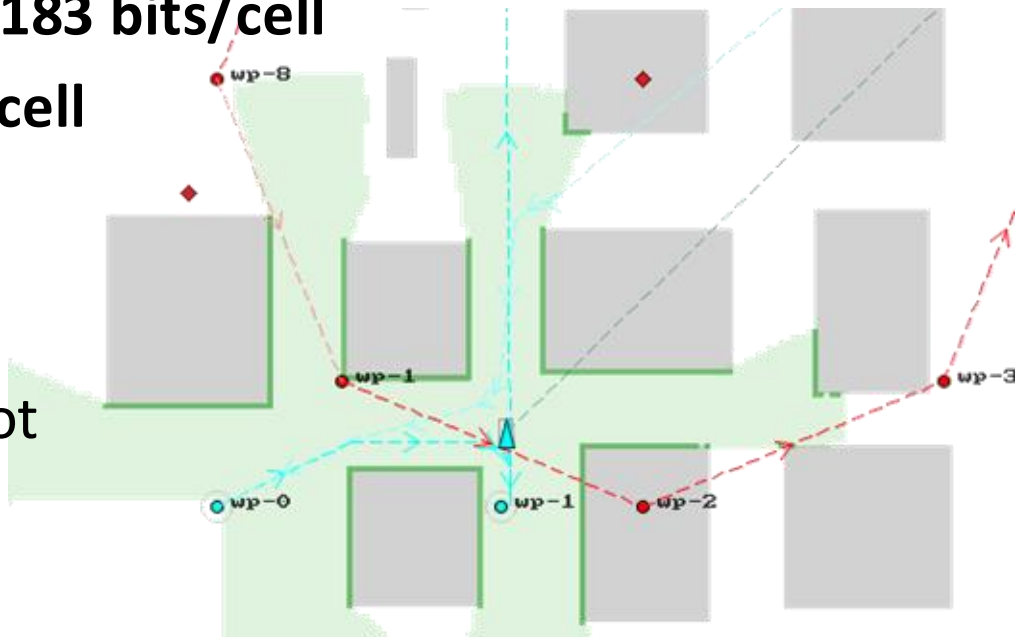
- If $D \ge p$, we can achieve $R(D) = 0$ by letting $\hat{X} = 0$ (with prob. 1)
  - In this case, $I(X;\hat{X}) = 0$ and $D = p$
- Similarly, if $D \ge 1-p$, we can achieve $R(D) = 0$ by setting $\hat{X} = 1$ (with probability 1)

$$R(D) = \begin{cases} H_{\mathrm{B}}(p) - H_{\mathrm{B}}(D), & 0 \le D \le \min\{p, 1-p\} \\ 0, & D > \min\{p, 1-p\} \end{cases}$$



Example: Robovac Binary Occupancy Grid

- A robovac builds a room occupancy grid where each grid cell is occupied (1) or free (0), with probability p=0.1 or 1-p=0.9
- It compresses and then sends the map to a remote server
- The distortion D is the fraction of cells reconstructed incorrectly
- **Error-free compression** requires $R(0)=H_B(0.1)\approx$**0.469 bits/cell**
- If we tolerate D**=0.05** wrong cells on average, then $R(0.05)= H_B(0.1)- H_B(0.05)\approx$**0.183 bits/cell**
- If $D \geq p$**=0.1**, then $R(D)$**=0 bits/cell**
  - I.e., the HQ could simply reconstruct every cell as free
- Sparse obstacle maps are cheap to compress if the robot can tolerate a small fraction of wrong cells



RD of Gaussian Source

- **Theorem:** The RD function for a RV $X \sim \mathcal{N}(0,\sigma^2)$ source with squared-error distortion is

$$R(D) = \begin{cases} \frac{1}{2}\log\frac{\sigma^2}{D}, & 0 \le D \le \sigma^2 \\ 0, & D > \sigma^2 \end{cases}$$

  - *Proof:* By the RD theorem extended to continuous alphabets, we have

$$R(D) = \min_{f(\hat{x}|x):\mathrm{E}\{(\hat{X}-X)^2\}\le D} I(X;\hat{X})$$

For any quantization method with limited distortion

As in the preceding example, we first find a lower bound for the rate distortion function and then prove that this is achievable

RD of Gaussian Source (proof cont'd)

- Since $\mathrm{E}\{(X-\hat{X})^2\} \le D$ *

Translation does not change DE (in average sense also)

$$I(X;\hat{X}) = h(X) - h(X|\hat{X}) = \frac{1}{2}\log(2\pi e\sigma^2) - h(X-\hat{X}|\hat{X})$$

Conditioning reduces DE (minus=increases)

Maxed by entropy of normal with same covariance

$$\ge \frac{1}{2}\log(2\pi e\sigma^2) - h(X-\hat{X}) \ge \frac{1}{2}\log(2\pi e\sigma^2) - h\left(\mathcal{N}\left(0, \mathrm{E}\left\{(X-\hat{X})^2\right\}\right)\right)$$

$$= \frac{1}{2}\log(2\pi e\sigma^2) - \frac{1}{2}\log\left(2\pi e\,\mathrm{E}\left\{(X-\hat{X})^2\right\}\right) \ge \frac{1}{2}\log(2\pi e\sigma^2) - \frac{1}{2}\log(2\pi eD)$$

Distortion is limited, per *

$$= \frac{1}{2}\log\frac{\sigma^2}{D} \qquad [\text{bits}]$$

- We used the fact that conditioning reduces entropy, and that the normal pdf maximizes the entropy for a given variance
- Hence

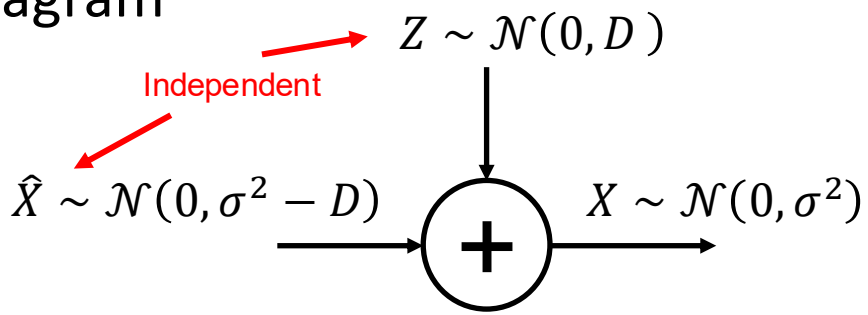
$$R(D) \ge \frac{1}{2}\log\frac{\sigma^2}{D}$$

RD of Gaussian Source (proof cont'd)

- To find the conditional pdf $f(\hat{x}|x)$ that achieves the lower bound, it is more convenient to look at the conditional pdf $f(x|\hat{x})$, also called the *test channel* (duality between RD and capacity)
  - We construct $f(x|\hat{x})$ to achieve equality in the bound
- We choose the joint pdf in the diagram
- If $D \le \sigma^2$, we choose

$$X = \hat{X} + Z, \qquad \hat{X} \sim \mathcal{N}(0, \sigma^2 - D)$$

$$Z \sim \mathcal{N}(0, D)$$



where $\hat{X}$ and Z are independent

- Achieves the bound, since $E\left\{\left(X - \hat{X}\right)^2\right\} = E\{Z^2\} = D$, and

$$I(X;\hat{X}) = h(X) - h(X|\hat{X}) = h(X) - h(X - \hat{X}|\hat{X}) = h(X) - h(Z|\hat{X}) = h(X) - h(Z)$$
$$= \frac{1}{2}\log(2\pi e\sigma^2) - \frac{1}{2}\log(2\pi eD) = \frac{1}{2}\log\frac{\sigma^2}{D}$$

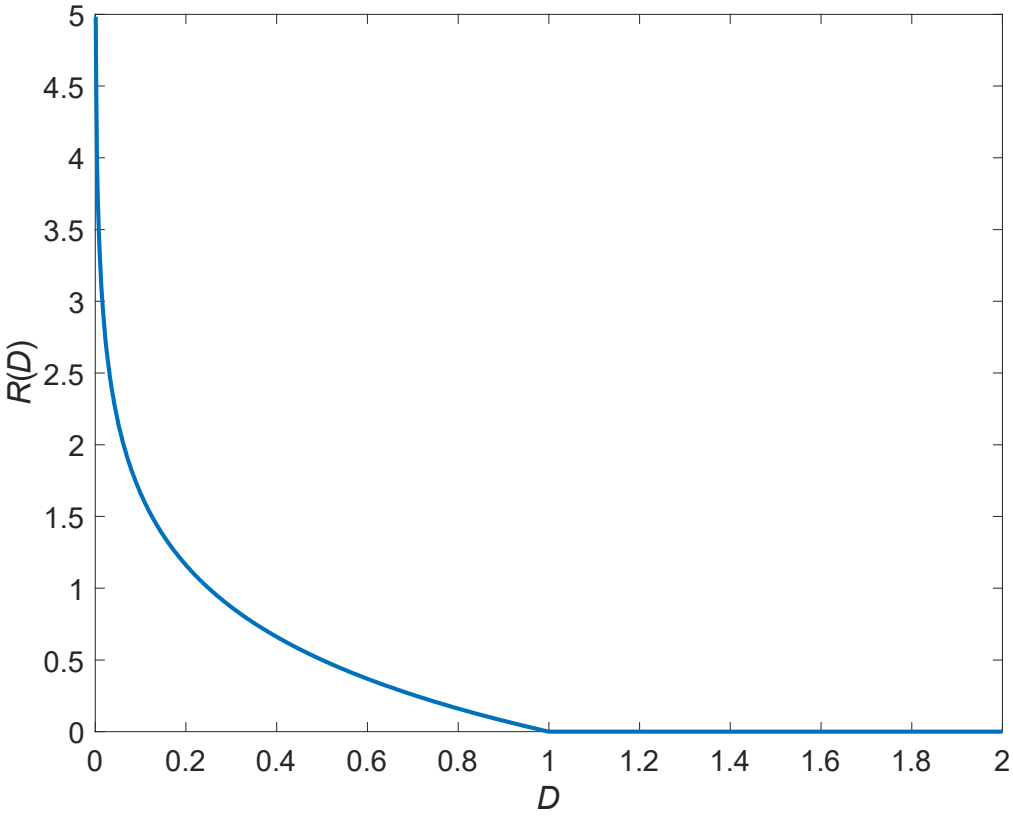
Z independent of $\hat{X}$

RD of Gaussian Source (proof cont'd)

- If $D \geq \sigma^2$, we choose $\hat{X} = 0$ (with probability 1), achieving $R(D) = 0$
- Hence, the RD function with squared-error distortion is

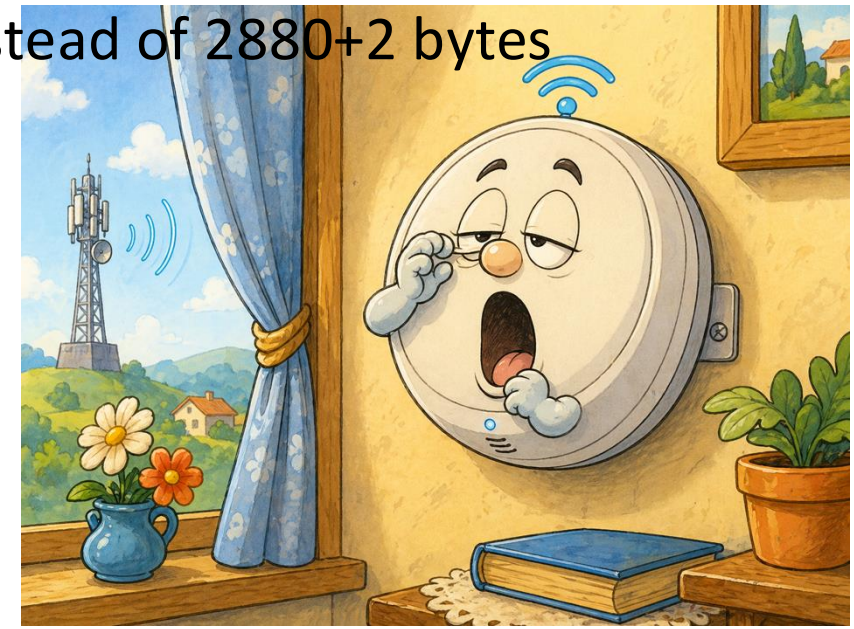
$$R(D) = \begin{cases} \frac{1}{2}\log\frac{\sigma^2}{D}, & 0 \leq D \leq \sigma^2 \\ 0, & D > \sigma^2 \end{cases}$$

- You can get $D=\sigma^2$ distortion without encoding any symbol (R=0), just reconstruct it with $\hat{X}$=0 (or the mean value)



Example: LoRaWAN Sensor Communication

- A battery-powered LoRaWAN sensor measures temperature every 1 min and sends compressed samples to the gateway
- The variance of the daily temperature (Gaussian) is $\sigma^2 = 1$
- Save daily mean, send the diff. with RMSE = 0.5C and D=0.5^2=0.25
- We obtain $R(D) = \frac{1}{2}\log\frac{\sigma^2}{D} = \frac{1}{2}\log\frac{1}{0.25} = 1$ bit/sample
- You send 180+2 bytes each day instead of 2880+2 bytes
  - Payload size reduced by 15.8x
  - If airtime accounts for 80% of energy consumption, battery life is extended from 1 to 4 years
- When $D \geq \sigma^2 = 1$ (RMSE=1 C), sending the daily mean is enough
  - You send only 2 bytes a day
  - Battery life extended up to 5 years



Distortion Rate of a Gaussian Source

- We can rewrite to express the distortion in terms of the rate

$$D(R) = \sigma^2 2^{-2R}$$

- Each bit of description reduces the distortion by 4 (6 dB/bit rule)
  - With a 1-bit description, the best expected square error is $\sigma^2/4=0.25\sigma^2$

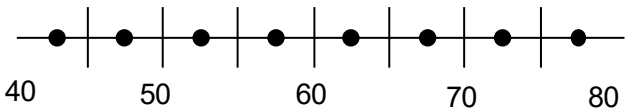
- However, remember the 1-bit quantization of a $\mathcal{N}(0,\sigma^2)$ RV
  - Using the two regions corresponding to the positive and negative real lines and reproduction points as the centroids of the respective regions, the expected distortion is $\sigma^2(\pi-2)/\pi = 0.363\sigma^2 > 0.25\sigma^2$
- In general, the $R(D)$ limit considers long blocks X^n
  - Achieve a lower distortion by considering several distortion problems in succession, instead of considering each problem separately
  - This is somewhat surprising because we are quantizing independent RVs!

Vector Quantization vs. Scalar Quantization

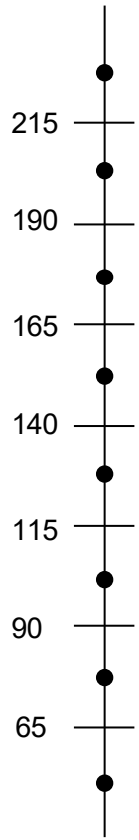
- We still need to understand why joint coding of source symbols can outperform separate coding
- The practical answer is **vector quantization**, as opposed to *scalar quantization*
  - For a given rate, vector quantization results achieves lower distortion than scalar quantization
- If the source output is correlated, source vectors values tend to cluster in space
  - This can be used to provide more efficient representations
- Even for uncorrelated source values, the vector quantizer is more efficient than the scalar quantizer
  - We have more design flexibility to match the quantizer to the source

Example: Vector Quantization of Correlated Sources

- Let height of individuals vary uniformly between 40 and 80 inches, and weight vary uniformly between 40 and 240 pounds
- We are given 6 total bits to represent each pair
  - We could use 3 bits for the height and 3 bits for the weight
- The weight range between 40 and 240 pounds would be divided into 8 equal intervals of 25 pounds, with reconstruction values {52,77,...,227}
- The height range between 40 and 80 inches can be divided into 8 intervals 5 inches wide, with reconstruction levels {42,47,...,77}
- Consider this quantization scheme in two dimensions



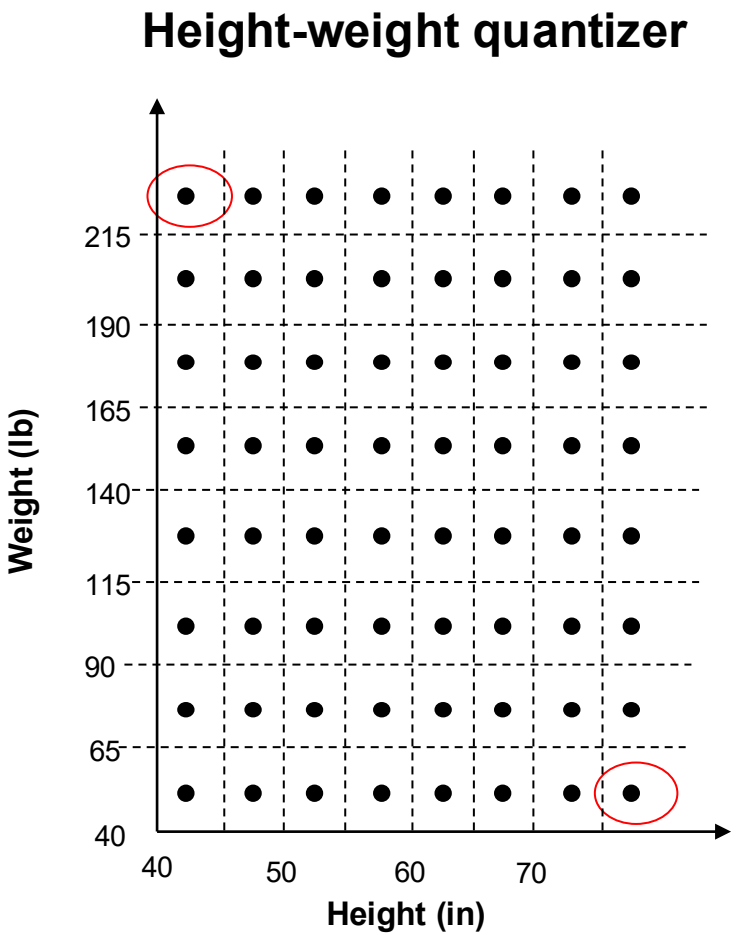
Height quantizer



Weight quantizer

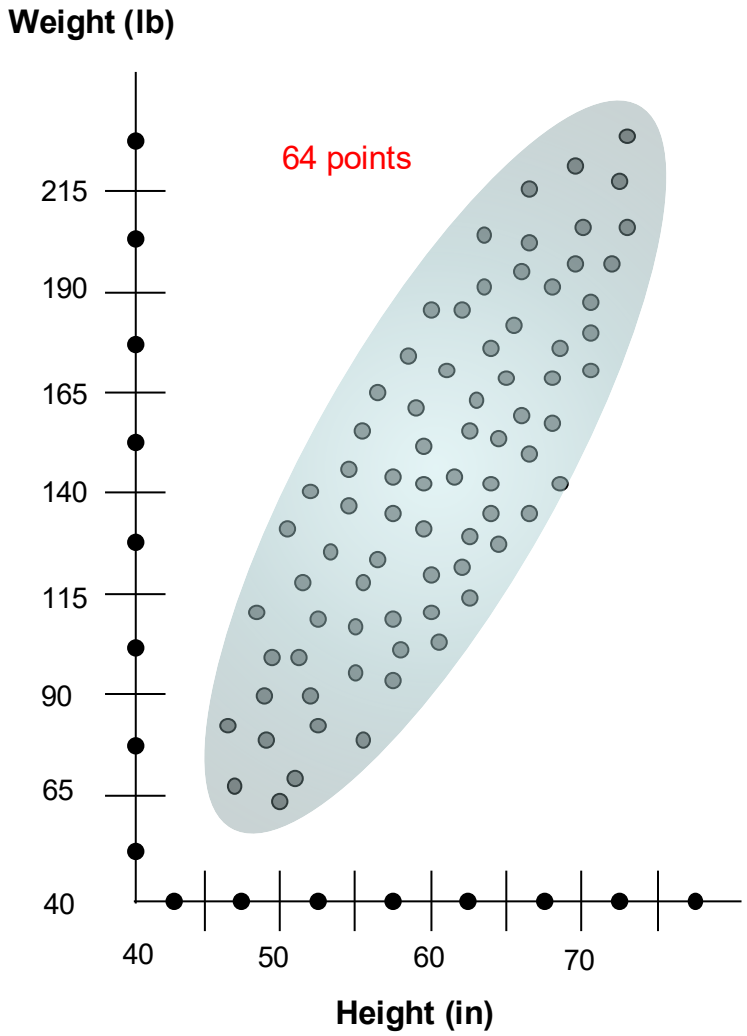
Example: Vector Quantization of Correlated Sources

- Same 8 height values and 8 weight values
- We have a point for
  - a person who is 80 inches tall and weighs 40 pounds
  - an individual whose height is 42 inches but weighs more than 200 pounds
- Obviously, these outputs will never be used
- Many reconstruction points are wasted, because they correspond to source vectors that never occur



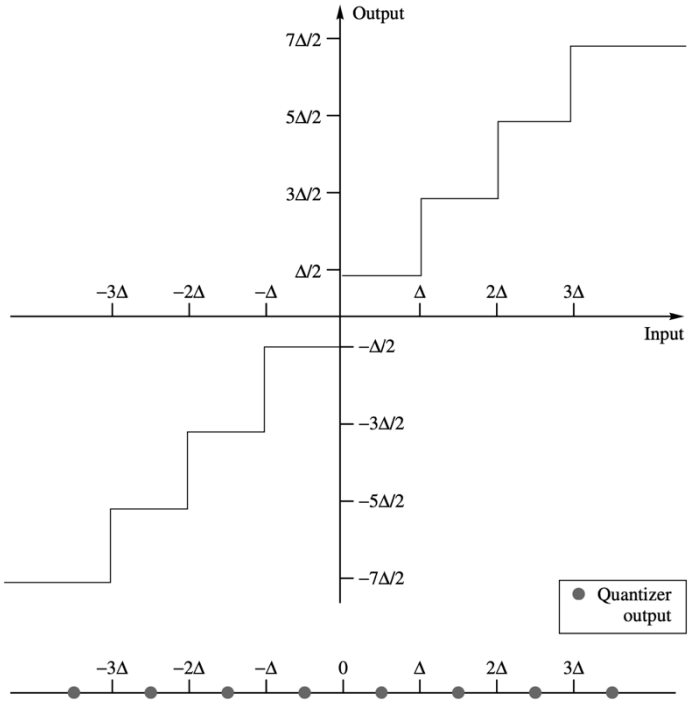
Example: Vector Quantization of Correlated Sources

- Consider the correlation between height and weight
- This quantizer has the same number of points, however, these are placed in the area more frequently occupied by the 2D input
- Using this quantizer, we can no longer quantize the height and weight separately
- We must treat height and weight jointly in 2D to find the closest quantizer output point
- This method provides a much better representation of the input



Why VQ Helps Even for Uncorrelated Sources

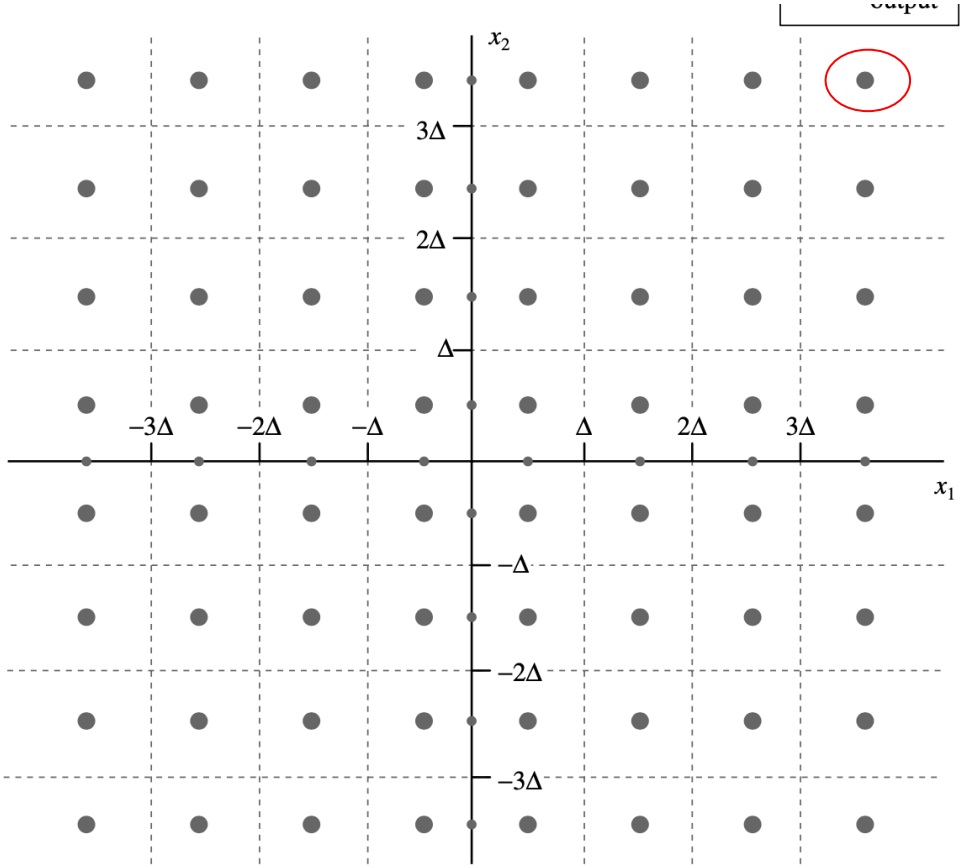
- Consider a Laplacian source quantized by an 8-level uniform scalar quantizer with $\Delta = 0.7309$
- Although the quantizer intervals are equally spaced, the source **does not** fall into them with equal probability
  - Input falls in $[0,\Delta)$ with probability 0.3242, and in $[3\Delta,\infty)$ with prob. 0.0225
- Therefore, some scalar outputs are used much more often than others
- When several independent Laplacian RVs are jointly quantized, a VQ can place more points in frequently used regions



Alphabet Size	Uniform		Gaussian		Laplacian	
	Step Size	SNR	Step Size	SNR	Step Size	SNR
2	1.732	6.02	1.596	4.40	1.414	3.00
4	0.866	12.04	0.9957	9.24	1.0873	7.05
6	0.577	15.58	0.7334	12.18	0.8707	9.56
8	0.433	18.06	0.5860	14.27	0.7309	11.39
10	0.346	20.02	0.4908	15.90	0.6334	12.81
12	0.289	21.60	0.4238	17.25	0.5613	13.98
14	0.247	22.94	0.3739	18.37	0.5055	14.98
16	0.217	24.08	0.3352	19.36	0.4609	15.84
32	0.108	30.10	0.1881	24.56	0.2799	20.46

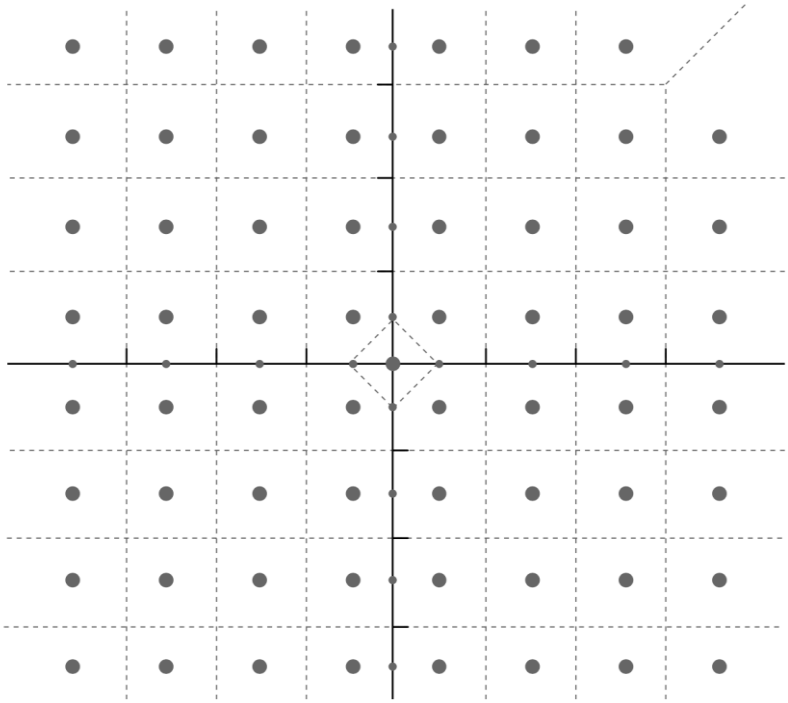
Example: Vector Quantization of Uncorrelated Sources

- Think in 2D the same quantization process of before
- The top rightmost point corresponds to 2 consecutive outputs with a value greater than 3Δ
- The probability of 2 consecutive outputs > 2.193 is 0.0225×0.0225 ≈ 0.0005
- Given that we this point is rarely used, it could be better placed in a high-probability region



Example: Vector Quantization of Uncorrelated Sources

- Move this rarely used reconstruction point to the origin
- If two consecutive samples are both larger than 3Δ, their joint reproduction need no longer be the Cartesian product of the two scalar reproduction values
- Comparing the two vector quantizers, the MSE for the 1st is about 0.3 dB higher than for the 2nd
- With larger input blocks, higher dimensions provide even more flexibility and potentially larger gains

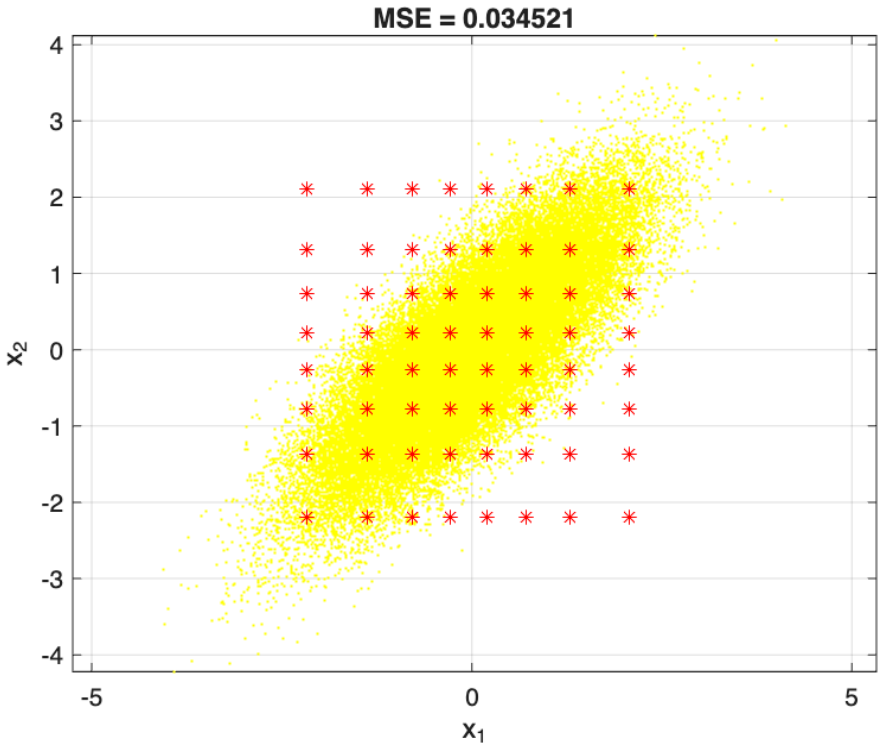


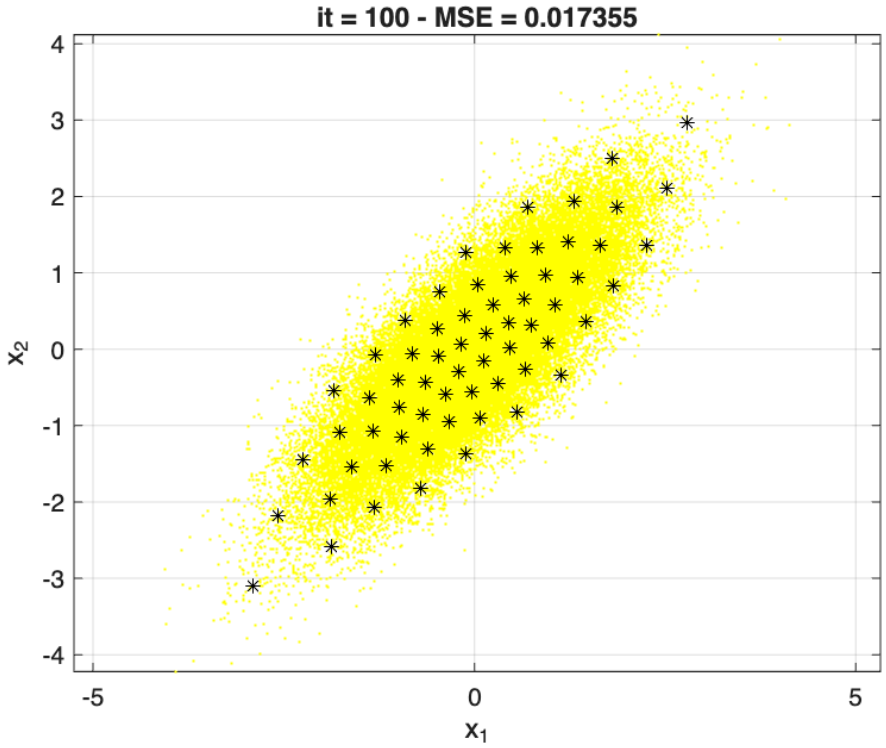
Linde-Buzo-Gray VQ Algorithm (1980)

- Given a training set of *N* vectors and a codebook with *K* vectors, we want to minimize the empirical average distortion *J*
- Each training vector is represented by its nearest codeword
- Try to place the *K* codewords where the source vectors lie
- LBG is a practical way to design a vector quantizer directly from training data, without explicit multidimensional integration
- LBG alternates two steps, as in Lloyd-Max
  - Nearest-neighbor assignment
  - Centroid update
- A common initialization is the LBG splitting method
  - Start from one centroid
  - Split each codeword into two nearby codewords
  - Run Lloyd iterations again
  - Repeat until the desired codebook size is reached

Example: LBG for Correlated Gaussian Data

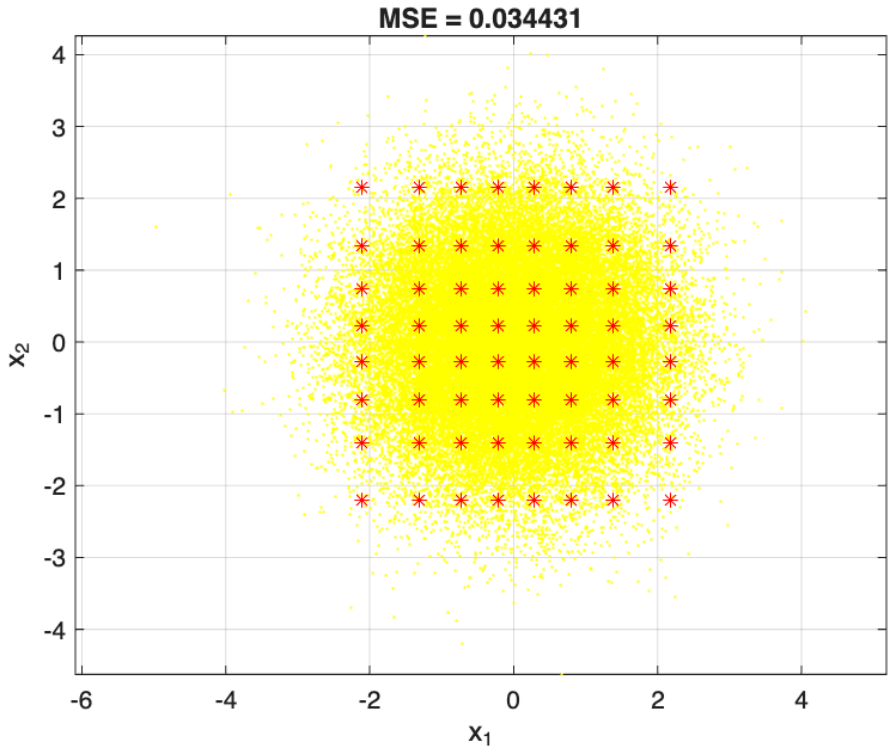
- Scalar quantize (left) separately two Gaussian components
- Strong correlation (ρ=0.8), many codewords are rarely used
- LBG (right) relocates the same 64 codewords along the cloud
- MSE decreases from 0.0345 to 0.0174 (3 dB) with no increase in rate

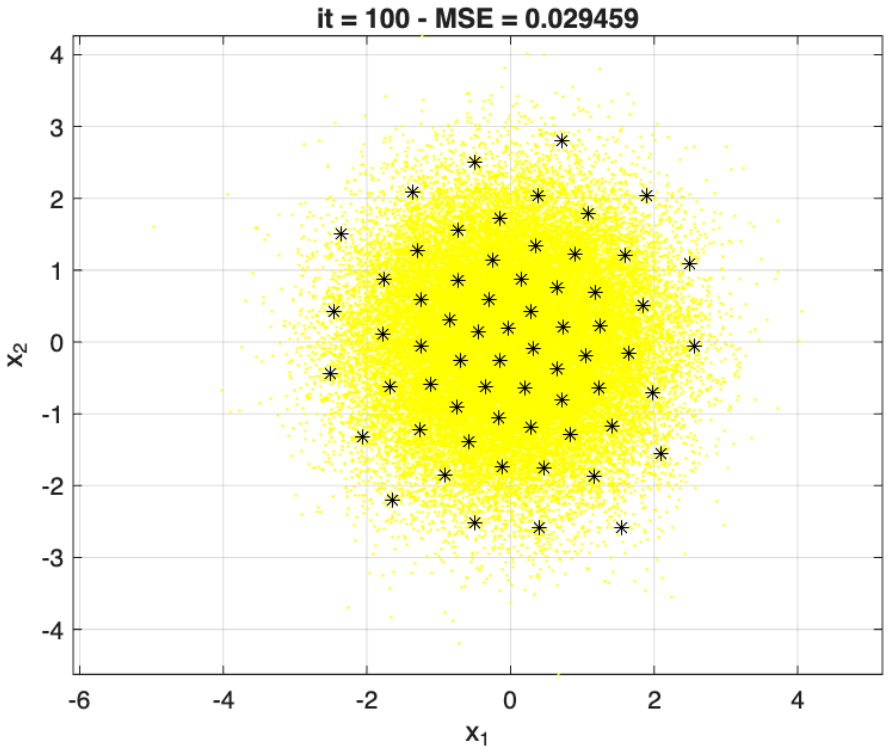




Example: LBG for Uncorrelated Gaussian Data

- No correlation (ρ=0), but LBG relocates the same 64 codewords toward the more populated region
- MSE anyway drops to 0.0295 (0.7 dB), still no increase in rate
- LBG tends to form an approximately hexagonal packing, which is more efficient than a rectangular grid for MSE quantization





From VQ to Parallel RV Coding

- Using vector quantization, we have shown that joint RV coding improves distortion even in the independent case
  - LBG moved codewords where they are useful
- For Gaussian sources, after decorrelating the vector with one of many techniques, we obtain independent RVs
- The problem then becomes to allocate the rate among independent Gaussian components
- This is the parallel **Gaussian source model**

RD of Parallel Gaussian Sources

- **Theorem:** Let $X_i \sim \mathcal{N}(0,\sigma^2_i)$, $i = 1,2,\ldots,m$, be independent Gaussian RVs, and let the distortion measure be $d(x^m,\hat{x}^m) = \sum_{i=1}^{m}(x_i - \hat{x}_i)^2$. Then the RD function is given by

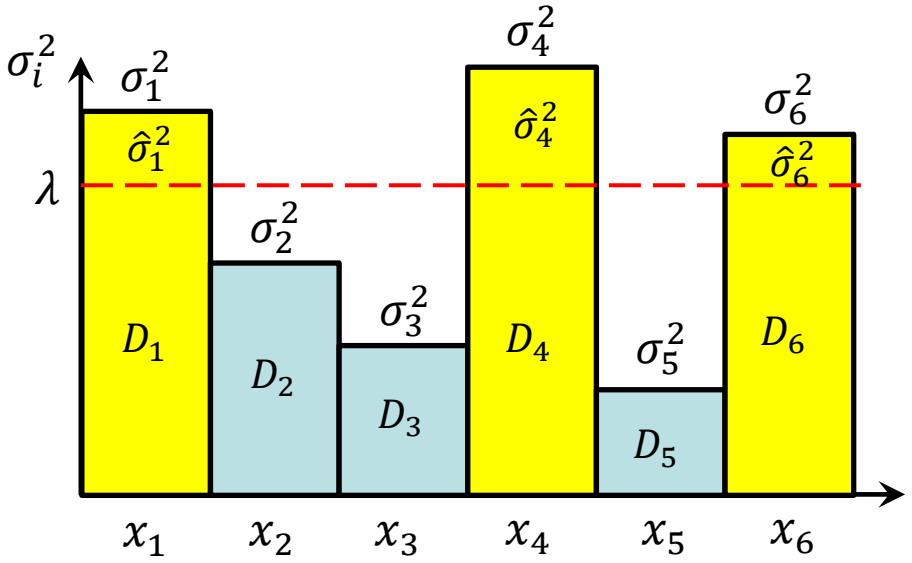
$$R(D) = \sum_{i=1}^{m} \frac{1}{2}\log\frac{\sigma_i^2}{D_i} = \sum_{i=1}^{m} \frac{1}{2}\max\left\{\log\frac{\sigma_i^2}{\lambda}, 0\right\}$$

where λ is chosen so that $\sum_{i=1}^{m} D_i = D$

Same residual distortion λ for all active RVs

$$D_i = \min\{\lambda, \sigma_i^2\} = \begin{cases} \lambda, & \lambda < \sigma_i^2 \\ \sigma_i^2, & \lambda \geq \sigma_i^2 \end{cases}$$

- Reverse water-filling
  - We choose a constant λ and allocate bits to those RVs with variances greater than λ
  - No bits are allocated to RVs with variance less than λ



RD of Parallel Gaussian Sources

- Summarizing, we start from the source X and arrive at description $\hat{X}$ characterized by

$$X^n \sim \mathcal{N}_n\left(0, \begin{bmatrix} \sigma_1^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_m^2 \end{bmatrix}\right) \quad \longrightarrow \hat{X}^n \sim \mathcal{N}_n\left(0, \begin{bmatrix} \hat{\sigma}_1^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \hat{\sigma}_m^2 \end{bmatrix}\right)$$

  where $D_i = \min\{\lambda, \sigma_i^2\} = \mathrm{E}\left\{\left(X_i - \hat{X}_i\right)^2\right\}$ is residual distortion

- The RD function for a multivariate normal (correlated RVs) can be obtained by reverse water-filling on the covariance eigenvalues
  - Perform eigen-decomposition of covariance matrix, $\mathbf{K} = \mathbf{U}\boldsymbol{\Gamma}\mathbf{U}^T$, where $\mathbf{U}$ is the $m \times m$ eigenvector matrix (each column), and $\boldsymbol{\Gamma} = \begin{pmatrix} \gamma_1^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \gamma_m^2 \end{pmatrix}$ is the diagonal eigenvalues matrix, with $\gamma_1^2 \geq \cdots \geq \gamma_m^2$
  - Then, $$D = \frac{1}{m}\sum_{i=1}^{m} \min\{\gamma_i^2, \lambda\}$$

RD of Parallel Gaussian Sources

- Finally, the RD is

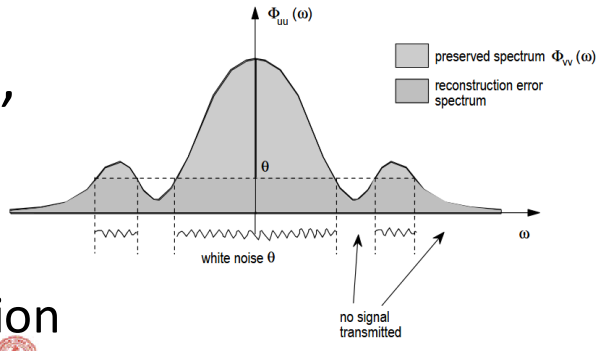
$$R(D) = \frac{1}{2}\sum_{i=1}^{m} \max\left\{0, \log\frac{\gamma_i^2}{\lambda}\right\}$$

  - For i.i.d. X^m

$$R(D) = \frac{m}{2}\log\frac{\sigma^2}{D}, \text{ or 0 if } D \geq \sigma^2$$

- Equivalent: apply the linear transformation $\mathbf{Y} = \mathbf{U}^T\mathbf{X}$ that decorrelates the *m* RVs, and proceed as usual

- Also, by the spectral representation theorem, a Gaussian SP is the integral of independent Gaussian SPs in the frequency bands
  - Reverse water-fill the spectrum to get the RD function
  - Replace sums with integrals



Convexity of RD Function

- **Lemma:** The rate $R(D)$ function is a nonincreasing convex function of distortion D
  - *Proof:* since $R(D)$ is the minimum of the mutual information over increasingly larger sets as D increases, then $R(D)$ is nonincreasing in D
    - ❖ Larger D means a looser distortion constraint, hence a larger feasible set
  - Consider two RD pairs (R_1,D_1) and (R_2,D_2) on the RD curve
  - Use the joint pdfs $p_1(x,\hat{x}) = p(x)p_1(\hat{x}|x)$ and $p_2(x,\hat{x}) = p(x)p_2(\hat{x}|x)$ to achieve
  - Consider the distribution $p_\lambda = \lambda p_1 + (1-\lambda)p_2$
  - Since the distortion is a linear function of the distribution, then $D_\lambda = \lambda D_1 + (1-\lambda)D_2$
  - Remember that mutual information is convex on the conditional pdf, then

R is the minimum

$$R(D_\lambda) \le I_{p_\lambda}(X;\hat{X})$$

Convex on p(x̂|x) for a fixed p(x)

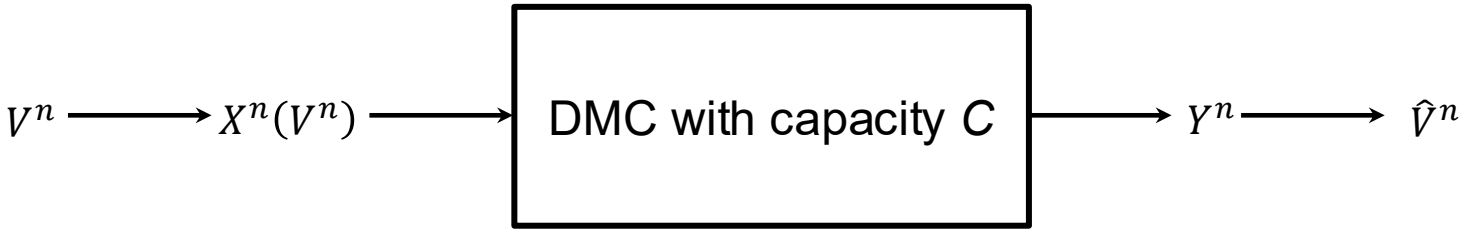
$$\le \lambda I_{p_1}(X;\hat{X}) + (1-\lambda) I_{p_2}(X;\hat{X})$$

Achieved pairs

$$= \lambda R(D_1) + (1-\lambda) R(D_2)$$

Source–Channel Separation with Distortion

- **Theorem:** Let $V_1, V_2, \ldots, V_n$ be a finite alphabet i.i.d. source which is encoded as a sequence of n input symbols X^n of a discrete memoryless channel with capacity C
- The output of the channel Y^n is mapped onto the reconstruction sequence $\hat{V}^n = g(Y^n)$
- Let $D = \mathrm{E}\{d(V^n, \hat{V}^n)\} = 1/n \sum_{i=1}^{n} \mathrm{E}\{d(V^i, \hat{V}^i)\}$ be the average distortion achieved by this combined source and channel coding scheme
- Then **distortion D is achievable iff $C > R(D)$**



Characterization of the RD Function

- We defined the information RD function as

$$R(D) = \min_{q(\hat{x}|x):\sum_{(x,\hat{x})} p(x)q(\hat{x}|x)d(x,\hat{x}) \le D} I(X;\hat{X})$$

  minimizing over all conditional PMFs $q(\hat{x}|x)$ for which the joint PMF $p(x)q(\hat{x}|x)$ satisfies the expected distortion constraint

- This is a convex minimization problem over all $q(\hat{x}|x) \ge 0$ satisfying $\Sigma_{\hat{x}} q(\hat{x}|x) = 1$ for all x, and $\Sigma_{x,\hat{x}} q(\hat{x}|x)p(x)d(x,\hat{x}) \le D$
- We use the Lagrange multipliers λ and $v(x)$ to find the solution

$I(X;\hat{X})$ — $r(\hat{x}) = \Sigma_x p(x)q(\hat{x}|x)$ is the PMF on $\hat{X}$ induced by $q(\hat{x}|x)$

$$J(q) = \sum_x \sum_{\hat{x}} p(x)q(\hat{x}|x) \log \frac{p(x)q(\hat{x}|x)}{p(x)r(\hat{x})} + \lambda \sum_x \sum_{\hat{x}} p(x)q(\hat{x}|x)d(x,\hat{x}) + \sum_x v(x) \sum_{\hat{x}} q(\hat{x}|x)$$

These equations are not easy to solve!

constraint that $q(\hat{x}|x)$ is a conditional PMF

Characterization of the RD Function

- A lemma allows us to rewrite the RD function as a double minimization

$$R(D) = \min_{r(\hat{x})} \; \min_{q(\hat{x}|x):\sum_{(x,\hat{x})} p(x)q(\hat{x}|x)d(x,\hat{x}) \le D} \sum_{x}\sum_{\hat{x}} p(x)q(\hat{x}|x)\log\frac{q(\hat{x}|x)}{r(\hat{x})}$$

where $r(\hat{x}) = \sum_x p(x)q(\hat{x}|x)$

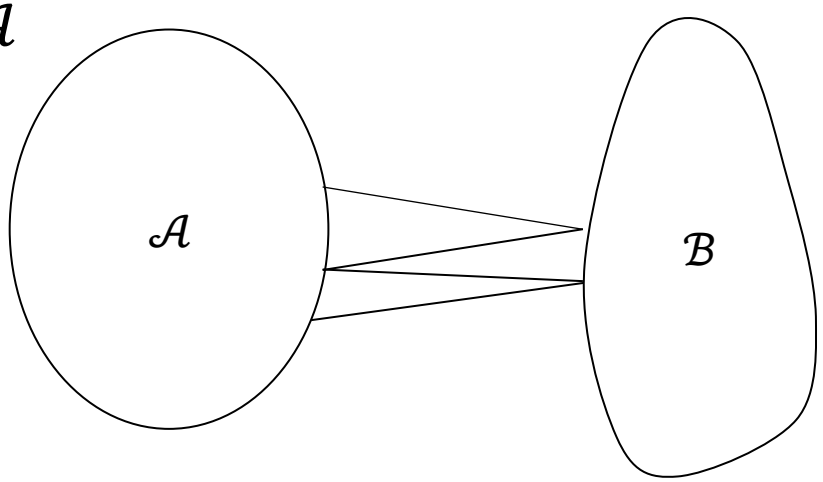
- If $\mathcal{A}$ is the set of all joint PMFs with marginal $p(x)$ that satisfy the distortion constraint and if $\mathcal{B}$ is the set of joint PMFs $q(\hat{x},x) = p(x)r(\hat{x})$ with arbitrary $r(\hat{x})$, we can see this as an **alternating minimization problem**

$$R(D) = \min_{q\in\mathcal{B}} \min_{p\in\mathcal{A}} D(p \parallel q)$$

The Alternating Minimization Algorithm

- Given two convex sets $\mathcal{A}$ and $\mathcal{B}$ in $\mathbb{R}^n$, find the minimum distance
- An intuitive algorithm takes Not all any point $x \in \mathcal{A}$ and finds its closest $y \in \mathcal{B}$
- Then, fix y and find the closest $x \in \mathcal{A}$
- Iterate to decrease this distance

$$d_{\min} = \min_{a \in \mathcal{A}, b \in \mathcal{B}} d(a, b)$$



- Csiszár and Tusnády have shown that for convex $\mathcal{A}$ and $\mathcal{B}$ and some conditions on $d(a,b)$, this **alternating minimization** algorithm will converge to the minimum
- If the sets are sets of probability distributions and the distance is the relative entropy, the algorithm converges to the **minimum relative entropy** between the two sets of distributions

The Blahut-Arimoto Algorithm

- Alternating minimization leads to **Blahut-Arimoto (BA) algorithm** in this context

1. Choose a λ multiplier and an initial PMF $r(\hat{x})$, and calculate the conditional PMF $q(\hat{x}|x)$ as

$$q(\hat{x}|x) = \frac{r(\hat{x})e^{-\lambda d(x,\hat{x})}}{\sum_{\hat{x}} r(\hat{x})e^{-\lambda d(x,\hat{x})}}$$

Normalization term

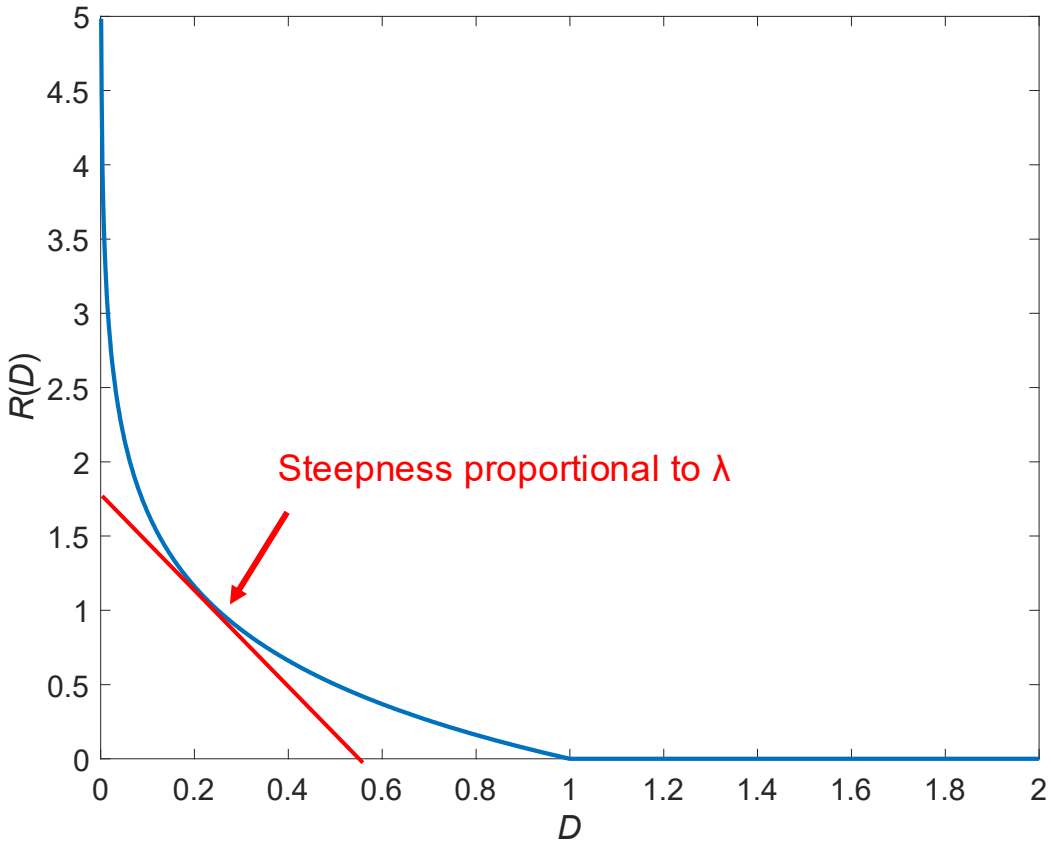
2. Then, for this PMF $q(\hat{x}|x)$, calculate the output PMF $r(\hat{x})$ as

$$r(\hat{x}) = \sum_{x} p(x)q(\hat{x}|x)$$

- Iteratively repeat steps 1-2: at each step, $D(p\|q)$ in the $R(D)$ double minimization equation is reduced
- BA deals only with the optimal partitioning of $\mathcal{X}$ given the representation $\widehat{\mathcal{X}}$, not with the optimal choice of $\widehat{\mathcal{X}}$

Characterization of the RD Function

- The limit of the BA algorithm is the $R(D)$ function, where D and $R(D)$ depend on λ
  - Choosing λ appropriately sweeps out the $R(D)$ curve
  - λ is proportional to the derivative of the $R(D)$ function wrt. D



Shannon Lower Bound on RD

- We can find a generic lower bound

$$R(D) = \min_{f(\hat{x}|x):E\{(\hat{X}-X)^2\}\le D} I(X;\hat{X}) = \min_{f(\hat{x}|x):E\{(\hat{X}-X)^2\}\le D} \left(h(X) - h(X|\hat{X})\right)$$

$$= h(X) - \max_{f(\hat{x}|x):E\{(\hat{X}-X)^2\}\le D} h(X|\hat{X}) = h(X) - \max_{f(\hat{x}|x):E\{(\hat{X}-X)^2\}\le D} h(X-\hat{X}|\hat{X})$$

Conditioning reduces entropy

$$\ge h(X) - \max_{f(\hat{x}|x):E\{(\hat{X}-X)^2\}\le D} h(X-\hat{X})$$

Entropy of translated RV

$$\ge \max\left\{h(X) - \frac{1}{2}\log 2\pi e D\,, 0\right\}$$

Gaussian has largest entropy for same variance

- Ideally, the source coder would produce distortions $X - \hat{X}$ that are statistically independent from the reconstructed signal

- For a fixed variance, the Gaussian RV is the hardest to compress: nonGaussian RVs have an R(D) curve not above the Gaussian one